Numerical Integration of Fourier Integrals by the Optimal Quadrature Formula Exact for Hyperbolic Functions

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Abstract. This paper studies the problem of construction of the optimal quadrature formula for numerical calculation of Fourier integrals. Applying the discrete analogue of the differential operator $\frac{d^4}{dx^4} - 2\frac{d^2}{dx^2} + 1$ and using its properties we obtain the optimal quadrature formula wich is exact for hyperbolic functions $\sinh(x)$ and $\cosh(x)$, we get explicit expressions for the coefficients of the optimal quadrature formula. The obtained optimal quadrature formulas can be used in problems where Fourier transformations are used.

INTRODUCTION

Numerical calculation of integrals of highly oscillating functions is one of the more important problems of numerical analysis, because such integrals are encountered in applications in many branches of mathematics as well as in other science. Standard methods of numerical integration frequently require more computational works and they cannot be successfully applied.

The earliest formulas for numerical integration of highly oscillatory functions were given by Filon [1] in 1928. The Filon's approach for Fourier integrals

$$I[f;\omega] = \int_a^b e^{2\pi i\omega x} f(x)dx,$$

is based on piecewise approximation of $f(x)$ by arcs of the parabola on the integration interval. Then finite integrals on the sub intervals are exactly integrated.

Afterwards for integrals with different type highly oscillating functions many special effective methods such as Filon-type method, Clenshaw-Curtis-Filon type method, Levin type methods, modified Clenshaw-Curtis method, generalized quadrature rule, Gauss-Laguerre quadrature are worked out (see, for example, [1, 2], for more review see, [3, 4] and references therein). In [5] the authors studied approximate computation of univariate oscillatory integrals for the standard Sobolev spaces H^s of periodic and non-periodic functions with an arbitrary integer $s \geq 1$. They found matching lower and upper bounds on the minimal worst case error of algorithms that use n function or derivative values. They also found sharp bounds on the information complexity which is the minimal n for which the absolute or normalized error is at most ε. In the work [6] the weighted lattice optimal cubature formulas in Sobolev's space $\widetilde{L_2^{(m)}}(\Omega)$ of periodic functions were constructed. In particular, from the result of the work [7], in univariate case when the weight is the function $\exp(i\sigma x)$ (where $x \in [0, 2\pi]$ and σ is an integer), the Babuška's optimal quadrature formula for Fourier coefficients [8**?**] was obtained.

In the work [9], using interpolation cubic spline with defect 1, the coefficients of quadrature formula for calculation of the Fourier integral are obtained. And, in [10], the optimal quadrature formula in the sense of Sard for calculation of the Fourier integrals was constructed in the space $L_2^{(2)}(0,1)$.

The present work is devoted to the construction of optimal quadrature formulas for the approximate calculation of Fourier integrals. Further, in the next section we discuss the statement of the problem.

STATEMENT OF THE PROBLEM

We consider the following quadrature formula

$$\int_0^1 e^{2\pi i\omega x} f(x)dx \cong \sum_{\beta=0}^{N} C_\beta f(h\beta) \tag{1}$$

with the error functional

$$\ell(x) = e^{2\pi i\omega x}\varepsilon_{[0,1]}(x) - \sum_{\beta=0}^{N} C_\beta \delta(x-h\beta), \tag{2}$$

where C_β are the coefficients of quadrature formula (1), $h = 1/N$, $N \in \mathbb{N}$, $i^2 = -1$, $\omega \in \mathbb{R}$ and $\omega \neq 0$, $\varepsilon_{[0,1]}(x)$ is the characteristic function of the interval $[0,1]$ and $\delta(x)$ is the Dirac delta-function.

Functions f are complex valued functions and belong to the Hilbert space $K_2(0,1)$, where

$$K_2(0,1) = \left\{ f : [0,1] \to \mathbb{C} \,|\, f' \text{ is absolutely continuous and } f'' \in L_2(0,1) \right\},$$

and in this space the inner product is defined by the equality

$$\langle f, g\rangle_{K_2} = \int_0^1 \Big(f''(x) - f(x)\Big)\Big(\overline{g}''(x) - \overline{g}(x)\Big)dx. \tag{3}$$

The norm of the function f is correspondingly defined by the formula

$$\|f\|_{K_2} = \langle f, f\rangle_{K_2}^{1/2}.$$

We note that the coefficients C_β depend on ω and N, i.e., $C_\beta = C_\beta(\omega, N)$.

The difference between the integral and the quadrature sum

$$(\ell, f) = \int_0^1 e^{2\pi\omega ix} f(x)dx - \sum_{\beta=0}^{N} C_\beta f(h\beta) = \int_{-\infty}^{\infty} \ell(x) f(x)dx \tag{4}$$

is called *the error* of the quadrature formula (1).

The error of the formula (1) is a linear functional in K_2^*, where K_2^* is the conjugate space to the space K_2.

According to the Cauchy-Schwarz inequality we have

$$|(\ell, f)| \leq \|f\|_{K_2} \cdot \|\ell\|_{K_2^*}.$$

Therefore, the error (4) of the formula (1) is estimated by the norm of the error functional (2)

$$\|\ell\|_{K_2^*} = \sup_{f,\ \|f|K_2\|=1} |(\ell, f)|.$$

Then, to estimate the error of the quadrature formula (1) over the functions of the space K_2 we find the norm of the error functional ℓ in the conjugate space K_2^*.

It is clear that the norm of the error functional ℓ depends on the coefficients C_β. The problem of finding the minimum of the norm of the error functional ℓ by coefficients C_β is called *Sard's problem*. And the obtained formula is called *the optimal quadrature formula.*

There are several methods for constructing optimal quadrature formulas such as the spline method, the ϕ-function method (see, for example [11], [12]) and the Sobolev method. Note that the Sobolev method is based on the construction of a discrete analogue to a linear differential operator (see, for instance [13, 14, 15]). In different spaces based on these methods, the construction problem of optimal quadrature formulas was studied by many scientists (see, for example, [1, 9, 11, 16, 17] and references therein).

The main aim of the present paper is to solve the Sard problem for quadrature formulas (1) in the space K_2 for any real ω exept zero, i.e., to find such coefficients $\mathring{C}_\beta$ that give the minimum to the norm of the error functional ℓ and calculate the quantity

$$\left\|\mathring{\ell}\right\|_{K_2^*} = \inf_{C_\beta} \|\ell\|_{K_2^*}. \tag{5}$$

Therefore, for constructing the optimal quadrature formula of the form (1) in the space K_2, we solve the following problems.

Problem 1. Calculate the norm of the error functional ℓ of the quadrature formula (1) in the space K_2^*.

Problem 2. Find the coefficients $\mathring{C}_\beta$ that give minimum to the norm of the error functional (2).

In the sequel of the paper we solve Problems 1 and 2 for the case $\omega \in \mathbb{R}$ and $\omega \neq 0$.

The rest of the paper is organized as follows. In section 3 we calculate the extremal function and using it we get the expression for the norm of the error functional (2). Section 4 presents definitions and auxiliary results. In Section 5 we get the explicit expression for optimal coefficients.

THE EXTREMAL FUNCTION AND THE NORM OF THE ERROR FUNCTIONAL (2)

To get the explicit expression for the norm of the error functional (2) in the space K_2^*, we use its extremal function. The function ψ_ℓ is called *the extremal function* for the functional ℓ (see, [14]), if the following equality holds

$$(\ell, \psi_\ell) = \|\ell\|_{K_2^*} \cdot \|\psi_\ell\|_{K_2}. \tag{6}$$

Since K_2 is a Hilbert space the extremal function ψ_ℓ in this space is found with the help of the general form of a linear continuous functional on Hilbert spaces given by the Riesz theorem. Then for the functional ℓ and for any $f \in K_2$ there exists a function $\psi_\ell \in K_2$ for which the following equation holds

$$(\ell, f) = \langle \psi_\ell, f \rangle_{K_2}, \tag{7}$$

where

$$\langle \psi_\ell, f \rangle_{K_2} = \int_0^1 \left(\psi_\ell''(x) - \psi_\ell(x)\right)\left(\overline{f}''(x) - \overline{f}(x)\right)dx \tag{8}$$

is the inner product of the functions ψ_ℓ and f defined in the space K_2.

From (7), taking into account (8), for the extremal function ψ_ℓ we get the following boundary value problem

$$\psi_\ell^{IV}(x) - 2\psi_\ell''(x) + \psi_\ell(x) = \ell(x), \tag{9}$$

$$\left(\psi_\ell^{(2+s)}(x) - \psi_\ell^{(s)}(x)\right)|_{x=0}^{x=1} = 0, \quad s = 0, 1. \tag{10}$$

For the solution of the boundary value problem (9)-(10) the following holds.

Theorem 1. The solution of the boundary value problem (9)-(10) is the extremal function ψ_ℓ of the error functional ℓ and it has the following form

$$\psi_\ell(x) = \int_0^1 e^{-2\pi i\omega y} G_2(x-y)dy - \sum_{\gamma=0}^{N} \overline{C_\gamma} G_2(x - h\gamma) + d_1 \sinh(x) + d_2 \cosh(x), \tag{11}$$

where

$$G_2(x) = \frac{\operatorname{sign}(x)}{4}(x\cosh(x) - \sinh(x)), \tag{12}$$

d_1 and d_2 are complex numbers.

Theorem 1 can be proved similarly as Theorem 2.1 of the work [18].

For the error functional (2) to be defined on the space K_2 it is necessary to impose the following conditions

$$(\ell, \sinh(x)) = 0, \tag{13}$$

$$(\ell, \cosh(x)) = 0. \tag{14}$$

The equalities (12) and (13) mean that the quadrature formula (1) is exact for $\sinh(x)$ and $\cosh(x)$.
Now, using Theorem 1 we get the expression for the norm of the error functional (2).
From (7), taking into account (2) and (11), we have

$$\begin{aligned}\|\ell\|^2_{K_2^*} &= (\ell, \psi_\ell) = \langle \psi_\ell, \psi_\ell \rangle_{K_2} \\ &= \sum_{\beta=0}^{N} \sum_{\gamma=0}^{N} C_\beta \overline{C_\gamma} G_2(h\beta - h\gamma) \\ &- \sum_{\beta=0}^{N} \int_0^1 (\overline{C_\beta} e^{2\pi i\omega x} + C_\beta e^{-2\pi i\omega x}) G_2(x - h\beta) dx \\ &+ \int_0^1 \int_0^1 e^{2\pi i\omega(x-y)} G_2(x-y) dy,\end{aligned} \tag{15}$$

where $\overline{C}_\beta$ is the complex conjugate of C_β. It is easy to show that the expression (14) is real.

Thus, Problem 1 is solved.

Further, we solve Problem 2. For this we need the following auxiliary results.

DEFINITIONS AND AUXILIARY RESULTS

In this section we give some definitions and known results that we need to prove the main results.

We use the concept of discrete argument functions and operations on them. The theory of discrete argument functions is given, for example, in [14, 15]. We give some definitions about functions of discrete argument.

Definition 1. The function $f(h\beta)$ is *a function of discrete argument* if it is given on some set of integer values of β.

Definition 2. *The inner product* of two discrete argument functions $f(h\beta)$ and $g(h\beta)$ is given by

$$[f(h\beta), g(h\beta)] = \sum_{\beta=-\infty}^{\infty} f(h\beta) \cdot g(h\beta),$$

if the series on the right-hand side of the last equality converges absolutely.

Definition 3. *The convolution* of two functions $f(h\beta)$ and $g(h\beta)$ is the inner product

$$f(h\beta) * g(h\beta) = [f(h\gamma), g(h\beta - h\gamma)] = \sum_{\gamma=-\infty}^{\infty} f(h\gamma) \cdot g(h\beta - h\gamma).$$

The Euler-Frobenius polynomials $E_k(x)$, $k = 1, 2, \ldots$ are defined by the following formula [15]

$$E_k(x) = \frac{(1-x)^{k+2}}{x} \left(x \frac{d}{dx} \right)^k \frac{x}{(1-x)^2}, \tag{16}$$

$E_0(x) = 1$.

For the Euler-Frobenius polynomials $E_k(x)$ the following identity holds

$$E_k(x) = x^k E_k\left(\frac{1}{x}\right). \tag{17}$$

In our computations we need the discrete analogue $D_2(h\beta)$ of the differential operator $\frac{d^4}{dx^4} - 2\frac{d^2}{dx^2} + 1$ which satisfies the following equality

$$D_2(h\beta) * G_2(h\beta) = \delta_d(h\beta), \tag{18}$$

where $G_2(h\beta) = \frac{\text{sign}(h\beta)}{4}\left((h\beta)\cosh(h\beta) - \sinh(h\beta)\right)$ is the discrete argument function corresponding to $G_2(x)$ defined by (12), $\delta_d(h\beta)$ is equal to 0 for $\beta \neq 0$ and it is equal to 1 for $\beta = 0$.

In [19] the discrete analogue $D_2(h\beta)$ of the differential operator $\frac{d^4}{dx^4} - 2\frac{d^2}{dx^2} + 1$, that satisfies equation (18) is constructed and the following was proved.

Theorem 2. The discrete analogue to the differential operator $\frac{d^4}{dx^4} - 2\frac{d^2}{dx^2} + 1$ has the form

$$D_2(h\beta) = \frac{2}{p}\begin{cases} A_1\lambda_1^{|\beta|-1} & \text{for } |\beta| \geq 2, \\ 1 + A_1 & \text{for } |\beta| = 1, \\ C_2 + \dfrac{A_1}{\lambda_1} & \text{for } \beta = 0, \end{cases} \tag{19}$$

where

$$\begin{aligned} p &= h\cosh(h) - \sinh(h), \\ p_1 &= \sinh(2h) - 2h, \\ A_1 &= \frac{(\lambda_1^2 + 1 - 2\lambda_1\cosh(h))^2}{\lambda_1^2 - 1}, \\ C_2 &= -4\cosh(h) + \frac{\lambda_1^2 + 1}{\lambda_1}, \lambda_1 = \frac{-p_1 + \sqrt{p_1^2 - 4p^2}}{2p} = \sqrt{3} - 2 + O(h^2), \end{aligned}$$

$|\lambda_1| < 1$ and h is a small positive parameter.

Properties of the discrete argument function $D_2(h\beta)$ were studied in [19]. Here we give the following properties of the discrete argument function $D_2(h\beta)$ which we use in our computations:

$$\begin{aligned} D_2(h\beta) * \sinh(h\beta) &= 0, \\ D_2(h\beta) * \cosh(h\beta) &= 0, \\ D_2(h\beta) * (h\beta)\sinh(h\beta) &= 0, \\ D_2(h\beta) * (h\beta)\cosh(h\beta) &= 0. \end{aligned} \tag{20}$$

COEFFICIENTS OF OPTIMAL QUADRATURE FORMULAS OF THE FORM (1)

To find the minimum of the expression (15) under the conditions (13) and (14) we apply the Lagrange method. Then we get the following system of linear equations for the coefficients C_β, $\beta = 0, 1, ..., N$, and d_1, d_2:

$$\sum_{\gamma=0}^{N} C_\gamma G_2(h\beta - h\gamma) + d_1\sinh(h\beta) + d_2\cosh(h\beta) = f_2(h\beta),\ \beta = 0, 1, ..., N, \tag{21}$$

$$\sum_{\gamma=0}^{N} C_\gamma \sinh(h\gamma) = q_s, \tag{22}$$

$$\sum_{\gamma=0}^{N} C_\gamma \cosh(h\gamma) = q_c, \tag{23}$$

where

$$f_2(h\beta)=\int_0^1 e^{2\pi i\omega x}G_2(x-h\beta)dx=\left[\frac{1}{(4\pi^2\omega^2+1)^2}\right]\cdot e^{2\pi i\omega h\beta}$$
$$-\frac{1}{4}\left[q_c-\frac{4\pi i\omega}{4\pi^2\omega^2+1}\right]\cdot(h\beta)\cosh(h\beta)+\frac{1}{4}\left[q_s+\frac{2}{4\pi^2\omega^2+1}\right]\cdot(h\beta)\sinh(h\beta)$$
$$+\frac{1}{4}\left[\frac{e^{2\pi i\omega}(4\pi^2\omega^2\cosh 1+(4\pi i\omega-1)\sinh 1)-2}{(4\pi^2\omega^2+1)^2}-\frac{e^{2\pi i\omega-1}}{2\pi i\omega-1}\right]\cosh(h\beta)$$
$$-\frac{1}{4}\left[\frac{e^{2\pi i\omega}(4\pi i\omega\cosh 1+(4\pi^2\omega^2-1)\sinh 1)+2\pi i\omega(4\pi^2\omega^2+3)}{(4\pi^2\omega^2+1)^2}+\frac{2e^{2\pi i\omega-1}}{(2\pi i\omega-1)^2}\right]\sinh(h\beta), \quad (24)$$

$$q_c=\frac{e^{2\pi i\omega}(\sinh 1-2\pi i\omega\cosh 1)+2\pi i\omega}{4\pi^2\omega^2+1}, \quad (25)$$

$$q_s=\frac{e^{2\pi i\omega}(\cosh 1-2\pi i\omega\sinh 1)-1}{4\pi^2\omega^2+1}. \quad (26)$$

Further, we solve the system (21)-(23). For this we assume that $C_\beta=0$ for $\beta=-1,-2,...$ and $\beta=N+1,N+2,...$. Then equalities (21) can be rewritten as

$$G_2(h\beta)*C_\beta+d_1\sinh(h\beta)+d_2\cosh(h\beta)=f_2(h\beta),\ \ \beta=0,1,2,...,N. \quad (27)$$

Now we can consider C_β as a discrete argument function.

We denote the left-hand side of (27) as a function of discrete argument

$$u(h\beta)=G_2(h\beta)*C_\beta+d_1\sinh(h\beta)+d_2\cosh(h\beta). \quad (28)$$

Then using discrete analogue $D_2(h\beta)$ given by (19) and its properties (20), we have

$$C_\beta=D_2(h\beta)*u(h\beta)\ \text{ for }\ \beta\in\mathbb{Z}. \quad (29)$$

Indeed,

$$\begin{aligned}D_2(h\beta)*u(h\beta)&=D_2(h\beta)*\big(G_2(h\beta)*C_\beta+d_1\sinh(h\beta)+d_2\cosh(h\beta)\big)\\&=(D_2(h\beta)*G_2(h\beta))*C_\beta=\delta_d(h\beta)*C_\beta=C_\beta.\end{aligned}$$

Thus, if we find the function $u(h\beta)$ then from equality (29) we can find C_β.

It is easy to see from (27) that

$$u(h\beta)=f_2(h\beta)\text{ for }\ \beta=0,1,2,...,N. \quad (30)$$

Now we find representation of the function $u(h\beta)$ for $\beta=-1,-2,...$ and $\beta=N+1,N+2,...$. For this we use the condition

$$C_\beta=0\text{ for }\ \beta=-1,-2,...\text{ and }\beta=N+1,N+2,...,N. \quad (31)$$

From (28), keeping in mind (30), using (25), (26), we get

$$u_2(h\beta)=\begin{cases}\frac{1}{4}\Big[(h\beta)\sinh(h\beta)-\cosh(h\beta)\Big]\cdot q_s+\frac{1}{4}\Big[-(h\beta)\cosh(h\beta)\\+\sinh(h\beta)\Big]\cdot q_c+\sinh(h\beta)\cdot d_1^-+\cosh(h\beta)\cdot d_2^-, & \beta\le 0,\\ f_2(h\beta), & 0\le\beta\le N,\\ \frac{1}{4}\Big[-(h\beta)\sinh(h\beta)+\cosh(h\beta)\Big]\cdot q_s+\frac{1}{4}\Big[(h\beta)\cosh(h\beta)-\sinh(h\beta)\Big]\cdot q_c\\+\sinh(h\beta)\cdot d_1^++\cosh(h\beta)\cdot d_2^+, & \beta\ge N,\end{cases} \quad (32)$$

where d_1^-, d_1^+, d_2^- and d_2^+ are unknowns. These unknowns can be found from (31). If we find d_1^-, d_1^+, d_2^- and d_2^+ then optimal coefficients $C_\beta,\ \beta = 0,1,...,N$ are found from (29).

In this work, we do not find d_1^-, d_1^+, d_2^- and d_2^+. Instead, using (29) we find the representation of optimal coefficients $C_\beta,\ \beta = 1,2,...,N-1$.

The following is valid.

Theorem 3. *The optimal coefficients satisfying the system (21)-(23) have the following form*

$$C_\beta = T(\omega)\cdot e^{2\pi i\omega h\beta} + m\cdot\lambda_1^\beta + n\cdot\lambda_1^{N-\beta},\ \beta = 1,2,...,N-1,$$

where $T(\omega), m$ and n are unknown.

Proof. From equation (29), taking into account equations (19) and (32), we arrive at the following when $\beta = 1,2,...,N-1$

$$C_\beta = D_2(h\beta) * u(h\beta) = D_2(h\beta) * f_2(h\beta) + m\cdot\lambda_1^\beta + n\cdot\lambda_1^{N-\beta}, \tag{33}$$

where

$$\begin{aligned} m = {} & \frac{2A_1}{\lambda_1 p}\sum_{\gamma=1}^{\infty}\lambda_1^\gamma\left(\frac{1}{4}(h\gamma)\sinh(h\gamma) - \frac{1}{4}\cosh(h\gamma)\right)\cdot q_s \\ & + \left(\frac{1}{4}(h\gamma)\cosh(h\gamma) - \frac{1}{4}\sinh(h\gamma)\right)\cdot q_c \\ & - d_1^-\sinh(h\gamma) + d_2^-\cosh(h\gamma) - f_2(-h\gamma), \end{aligned} \tag{34}$$

$$\begin{aligned} n = {} & \frac{2A_1}{\lambda_1 p}\sum_{\gamma=1}^{\infty}\lambda_1^\gamma\left(-\frac{1}{4}(1+h\gamma)\sinh(1+h\gamma) - \frac{1}{4}\cosh(1+h\gamma)\right)\cdot q_s \\ & + \left(\frac{1}{4}(1+h\gamma)\cosh(1+h\gamma) - \frac{1}{4}\sinh(1+h\gamma)\right)\cdot q_c \\ & + d_1^+\sinh(1+h\gamma) + d_2^+\cosh(1+h\gamma) - f_2(1+h\gamma), \end{aligned} \tag{35}$$

Now we calculate the convolution $D_2(h\beta) * f_2(h\beta)$ in (33).

We use properties (20) of the discrete argument function $D_2(h\beta)$ to calculate this convolution. In this case

$$D_2(h\beta) * f_2(h\beta) = D_2(h\beta) * \frac{1}{(4\pi^2\omega^2+1)^2}e^{2\pi i\omega h\beta}.$$

From here

$$D_2(h\beta) * f_2(h\beta) = T(\omega)\cdot e^{2\pi i\omega h\beta}. \tag{36}$$

Theorem 3 is proved.

Remark 1. The quantity $T(\omega)$ in equation (36) can also be found by direct calculation. But we find this quantity by another way below.

The main result of this work is as follows.

Theorem 4. There is a unique optimal quadrature formula of the form

$$\int_0^1 e^{2\pi i\omega x} f(x)dx \cong \sum_{\beta=0}^{N} C_\beta f(h\beta)$$

in the K_2 Hilbert space, and its coefficients are expressed as follows

$$\begin{aligned} C_0 = {} & \frac{1}{\sinh 1}((\sinh 1\cdot q_c - \cosh 1\cdot q_s) + T(\omega)(\cosh 1\cdot a_1 - \sinh 1\cdot a_4) \\ & + m(\cosh 1\cdot(a_2+a_3) - \sinh 1\cdot(a_5+a_6))) \\ C_\beta = {} & T(\omega)\cdot e^{2\pi i\omega h\beta} + m\left(\lambda_1^\beta + \cdot\lambda_1^{N-\beta}\right),\ \beta = 1,2,...,N-1, \\ C_N = {} & \frac{1}{\sinh 1}\left(q_s - (T(\omega)\cdot a_1 + m(a_2+a_3))\right), \end{aligned}$$

where

$$a_1 = \frac{e^{2\pi i\omega h(1+N)}\sinh(1-h) - e^{2\pi i\omega}\sinh 1 + e^{2\pi i\omega h}\sinh(h)}{e^{4\pi i\omega h} - 2\cosh(h)e^{2\pi i\omega h} + 1},$$

$$a_2 = \frac{1}{2}\left(\frac{\lambda_1^N e - \lambda_1 e^h}{\lambda_1 e^h - 1} - \frac{\lambda_1^N e^{h-1} - \lambda_1}{\lambda_1 - e^h}\right),$$

$$a_3 = \frac{1}{2}\left(\frac{\lambda_1 e^{h-1} - \lambda_1^N}{\lambda_1 e^h - 1} - \frac{\lambda_1 e - \lambda_1^N e^h}{\lambda_1 - e^h}\right),$$

$$a_4 = \frac{e^{2\pi i\omega h(1+N)}\cosh(1-h) - e^{4\pi i\omega h} - e^{2\pi i\omega h}\cosh 1 + e^{2\pi i\omega h}\cosh(h)}{e^{4\pi i\omega h} - 2\cosh(h)e^{2\pi i\omega h} + 1}$$

$$a_5 = \frac{1}{2}\left(\frac{\lambda_1^N e - \lambda_1 e^h}{\lambda_1 e^h - 1} + \frac{\lambda_1^N e^{h-1} - \lambda_1}{\lambda_1 - e^h}\right),$$

$$a_6 = \frac{1}{2}\left(\frac{\lambda_1^N e^h - \lambda_1 e}{\lambda_1 - e^h} - \frac{\lambda_1 e^{h-1} - \lambda_1^N}{\lambda_1 e^h - 1}\right),$$

$$T(\omega) = \frac{2\cdot(e^{4\pi i\omega h} - 2\cosh(h)\cdot e^{2\pi i\omega h} + 1)^2}{(4\pi^2\omega^2+1)^2 e^{2\pi i\omega h}\cdot P_2(e^{2\pi i\omega h})}, \tag{37}$$

$$m = \frac{2(\lambda_1 - e^h)(\lambda_1^N e^{2\pi i\omega} - 2\lambda_1^{2N} + 1)(\cos(2\pi\omega h) - \cosh(h) + \frac{T}{2}(4\pi^2\omega^2+1)\sinh(h))(\lambda_1 e^h - 1)}{\lambda_1(4\pi^2\omega^2+1)(\cos(2\pi\omega h))(e^{2h}-1)(\lambda_1^{2N}-1)}, \tag{38}$$

λ_1 is given in Theorem 1.

Proof. First, for the coefficients C_0 and C_N, calculating from (22) and (23), we obtain the following

$$\begin{aligned} C_0 = {} & \frac{2\pi i\omega\sinh 1 - e^{2\pi i\omega} + \cosh 1}{(4\pi^2\omega^2+1)\sinh 1} - \frac{T(\omega)}{2}\cdot\frac{\sinh(h)e^{2\pi i\omega} + \sinh(1-h) - \sinh 1\cdot e^{2\pi i\omega h}}{(\cos(2\pi\omega h) - \cosh(h))\sinh 1} \\ & + \frac{m}{2}\cdot\frac{-\lambda_1^{N+1}(e^{2h}-1) + \lambda_1(e^{2h-1} - e) + 2\lambda_1^2 e^h\sinh 1}{(\lambda_1 e^h - 1)(\lambda_1 - e^h)\sinh 1} \\ & + \frac{n}{2}\cdot\frac{\lambda_1^{N+1}(e^{2h-1} - e) - \lambda_1(e^{2h}-1) + 2\lambda_1^N e^h\sinh 1}{(\lambda_1 e^h - 1)(\lambda_1 - e^h)\sinh 1}, \end{aligned} \tag{39}$$

$$\begin{aligned} C_N = {} & \frac{e^{2\pi i\omega}(\cosh 1 - 2\pi i\omega\sinh 1) - 1}{(4\pi^2\omega^2+1)\sinh 1} - \frac{T(\omega)}{2}\cdot\frac{e^{2\pi i\omega h(1+N)}\sinh(1-h) - e^{2\pi i\omega}\sinh 1 + e^{2\pi i\omega h}\sinh(h)}{e^{2\pi i\omega h}(\cos(2\pi\omega h) - \cosh(h))\sinh 1} \\ & + \frac{m}{2}\cdot\frac{\lambda_1^{N+1}e(1-e^{2h-2}) - \lambda_1 e^h(e+e^h) - \lambda_1}{(\lambda_1 e^h - 1)(\lambda_1 - e^h)} \\ & + \frac{n}{2}\cdot\frac{2\lambda_1^2 e^h\cosh 1 - \lambda_1 e(e^{2h-2} - 1) + \lambda_1^{N+1}(e^{2h}-1)}{(\lambda_1 e^h - 1)(\lambda_1 - e^h)}. \end{aligned} \tag{40}$$

Now, from (21), taking into account (24),(39) and (40), we get expressions for are $T(\omega)$, m and n. From these expressions we get $m = n$ and equalities (37) and (38).

Theorem 4 is proved.

CONCLUSION

In this paper, we studied the problem of construction of the optimal quadrature formula in the form (1) for numerical calculation of Fourier integrals with $\omega \in R, \omega \neq 0$. We found the coefficients of the optimal quadrature formula are found. We note that the obtained optimal quadrature formula in the K_2 space is exact for hyperbolic sine and hyperbolic cosine functions.

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